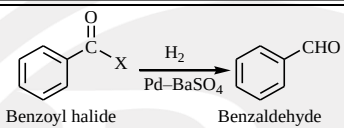
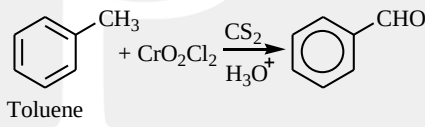
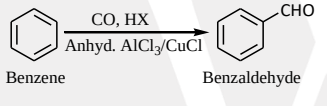
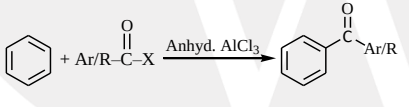


CHAPTER

6

Aldehydes, Ketones and Carboxylic Acids

Sr. No.	Concept	Description	Other information
1.	Name reactions:		
	(a) Rosenmund reduction	 Benzoyl halide $\xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2}$ Benzaldehyde	X = Cl Reagent = H ₂ , Pd-BaSO ₄
	(b) Stephen reaction	$\text{RCN} + \text{SnX}_2 + \text{HX} \rightarrow \text{RCH}=\text{NH} \xrightarrow{\text{H}_3\text{O}^+} \text{RCHO}$	X = Cl
	(c) Etard reaction	 Toluene $\xrightarrow[\text{H}_3\text{O}^+]{\text{CS}_2, \text{CrO}_2\text{Cl}_2}$ Benzaldehyde	Reagent = CS ₂ , H ₃ O ⁺
	(d) Gatterman-Koch reaction	 Benzene $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HX}}$ Benzaldehyde	X = Cl Reagent = Anhyd. AlCl ₃ /CuCl
	(e) Friedal-Crafts Acylation reaction	 Benzene + Ar/R-CO-X $\xrightarrow{\text{Anhyd. AlCl}_3}$ Acylbenzene	X = Cl, Br, I
	(f) Clemmensen reduction	$\text{C}=\text{O} \xrightarrow[\text{HCl}]{\text{Zn-Hg}} \text{CH}_2 + \text{H}_2\text{O}$	Reagent = Zn-Hg, HCl
	(g) Wolff Kishner reduction	$\text{C}=\text{O} \xrightarrow[-\text{H}_2\text{O}]{\text{NH}_2\text{NH}_2} \text{C}=\text{NNH}_2 \xrightarrow[\text{heat}]{\text{KOH, ethylene glycol}} \text{CH}_2 + \text{N}_2$	Reagent = NH ₂ NH ₂ , KOH, ethylene glycol
	(h) Haloform reaction	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 \xrightarrow{\text{NaOX}} \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa} + \text{CHX}_3$	X = Cl, Br, I Reagent = NaOX
	(i) Aldol condensation	$2 \text{R}-\text{CHO} \xrightleftharpoons{\text{dil. NaOH}} \text{R}-\text{CH}=\text{CH}-\text{CHO}$	Reagent = dil. NaOH R = Alkyl group Carbonyl group must have 'α' hydrogen.

Sr. No.	Concept	Description	Other information
	(j) Cannizzaro reaction (k) Decarboxylation (l) Hell-Volhard-Zelinsky reaction	$ \begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{O} + \text{C}=\text{O} + \text{Conc. KOH} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \\ \text{Formaldehyde} \end{array} $ $ \xrightarrow{\Delta} \begin{array}{c} \text{H} \quad \text{O} \\ \quad // \\ \text{H}-\text{C}-\text{OH} + \text{H}-\text{C} \\ \quad \\ \text{H} \quad \text{OK} \\ \text{Methanol} \quad \text{Potassium formate} \end{array} $ $ \text{R}-\text{COONa} \xrightarrow[\text{Heat}]{\text{NaOH \& CaO}} \text{R}-\text{H} + \text{Na}_2\text{CO}_3 $ $ \text{R}-\text{CH}_2-\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red phosphorus}} \text{R}-\underset{\text{X}}{\text{CH}}-\text{COOH} $ α-Halocarboxylic acid	Reagent = Conc. KOH Carbonyl groups having no ' α ' hydrogen atoms are taken. Reagent = NaOH & CaO X = Cl, Br Reagent = X ₂ /Red phosphorus, H ₂ O
2.	Tollen's Test	$\text{RCHO} + 2[\text{Ag}(\text{NH}_3)_2]^+ + 3\text{OH}^- \rightarrow \text{RCOO}^- + 2\text{Ag} + 2\text{H}_2\text{O} + 4\text{NH}_3$	Silver mirror is formed with Aldehyde
3.	Fehling's Test	$\text{R}-\text{CHO} + 2\text{Cu}^{2+} + 5\text{OH}^- \rightarrow \text{RCOO}^- + \text{Cu}_2\text{O} + 3\text{H}_2\text{O}$	Red ppt is formed with Aldehyde
4.	Distinction between phenol and carboxylic acid (a) NaHCO ₃ Test (b) FeCl ₃ Test	$ \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} \xrightarrow{\text{aq. NaHCO}_3} \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa}^+ + \text{CO}_2(\text{g}) $ No reaction with carboxylic acid.	Brisk effervescence of CO ₂ gas with carboxylic acid Violet colour with phenol
5.	Acidity order of Carboxylic acid (a) Effect of these groups in increasing acidity order (b) Increasing order of acidity	$\text{Ph} < \text{I} < \text{Br} < \text{Cl} < \text{F} < \text{CN} < \text{NO}_2 < \text{CF}_3$ $ \text{CF}_3\text{COOH} > \text{CCl}_3\text{COOH} > \text{CHCl}_2\text{COOH} > \text{NO}_2\text{CH}_2\text{COOH} > \text{NC-CH}_2\text{COOH} > \text{FCH}_2\text{COOH} > \text{CICH}_2\text{COOH} > \text{BrCH}_2\text{COOH} > \text{HCOOH} > \text{ClCH}_2\text{CH}_2\text{COOH} > \text{C}_6\text{H}_5\text{COOH} > \text{C}_6\text{H}_5\text{CH}_2\text{COOH} > \text{CH}_3\text{COOH} > \text{CH}_3\text{CH}_2\text{COOH} $ (continue) (continue)	Acidity $\propto$ EWG attached Acidity $\propto \frac{1}{+I \text{ effect}}$

Sr. No.	Concept	Description	Other information
6.	Physical properties of aldehydes & ketones (a) Physical state (b) Boiling point (c) Solubility	(i) Upto $C_{10} \rightarrow$ colorless liquid. (ii) Higher than $C_{10} \rightarrow$ solid with fruity order $HCHO < RCHO < RCOR$ (i) $R-CHO > R-CH_2-CHO$ (ii) $R-\overset{\overset{O}{\parallel}}{C}-R' > R-CH_2-\overset{\overset{O}{\parallel}}{C}-R'$	B.P. $\propto$ Mwt. Solubility $\propto \frac{1}{\text{alkyl chain}}$
7.	Physical properties of Carboxylic acids (a) Physical state (b) Boiling point (c) Solubility	(i) Upto $C_9 \rightarrow$ colorless liquid (ii) Higher than $C_9 \rightarrow$ colorless waxy solids $RCH_2COOH < R_2CHCOOH < R_3CCOOH$ $R_3C-COOH < R_2CHCOOH < RCH_2COOH$	B.P. $\propto$ Mwt. B.P. $\propto$ H-bonding Solubility $\propto \frac{1}{\text{No. of carbons atoms}}$ R = alkyl group
8.	Oxidising agents (a) Alkaline $KMnO_4$ (b) Jones reagent	$RCH_2OH \xrightarrow[H_3O^+]{\text{alkaline } KMnO_4} RCOOH$ $RCH_2OH \xrightarrow{CrO_3 \cdot H_2SO_4} RCOOH$	
9.	Reducing agents (a) $LiAlH_4$ and $NaBH_4$	(i) Aldehyde $\xrightarrow{R.A.} 1^\circ OH$ (ii) Ketone $\xrightarrow{R.A.} 2^\circ OH$ (iii) Carboxylic acid $\xrightarrow{LiAlH_4} 1^\circ OH$	R.A. (Reducing agent) $LiAlH_4$ and $NaBH_4$